

BARCELONA, SPAIN • ARCHITECTURAL MONOGRAPH



BIMCO

VOLUME I: LUXURY VILLAS & RESIDENCES

A DIALOGUE BETWEEN TOPOGRAPHY, FORM & BUILD INTEGRITY

Curated Collection of Sloped Villas, Desert Retreats & Executive Construction Records

COLLABORATIVE LEADERSHIP & PRINCIPALS

SOHEIL MASTI & SIAVASH PAZOOKI

BIM Director & Computational Architect | Senior Architectural Designer & Visualizer

BIMCO STUDIO • BARCELONA, SPAIN

BARCELONA, SPAIN • EDITION 2026

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SPAIN

THE PHILOSOPHY OF THE VILLA

Form, Topography, and the Boundary Between Inside & Outside

A private residence is the purest expression of architectural poetry and structural integrity. At BIMCO Studio Barcelona, our villa designs do not impose themselves upon the landscape; they grow organically from the earth's natural contours. Whether carving into steep forest hillsides or framing panoramic desert horizons, each residence is conceived through a harmonious balance of raw materiality—exposed concrete, tactile timber, natural stone—and endless natural daylight.

Every villa in this volume showcases complete architectural documentation: from initial hand sketches and parametric BIM models to active on-site construction supervision.

VILLA PROJECTS SCENARIO & INDEX

Structured Progression from Mountain Slopes to Built Realities

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ARCHITECTURAL BRIEF & SPATIAL CONCEPT

From Barcelona, BIMCO redefines luxury residential design through harmonious integrat...

BIM & STRUCTURAL INTEGRITY SPECS

Revit BIM LOD 350, Sloped Topography Mesh, Parametric Sun/Shadow Studies, Corona...

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Section B - B

Sc 1:100



Section C-C

Sc 1:100

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Left: Complete architectural floor plans & sections. Right: Signature organic forest ...



BIM & STRUCTURAL INTEGRITY SPECS

Topographic contour integration, thermal mass walls, environmental site drainage...

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ترکیب بتن، شیشه و طبیعت

این ویلا با الهام از توپوگرافی زمین و در هماهنگی کامل با طبیعت اطراف، در فضایی آرام و خصوصی طراحی شده است. خطوط ساده، متربال‌های طبیعی و دید باز به کوهستان، احساس آرامش و رفاه را در کنار زیبایی معماری مدرن به ارمغان می‌آورد.



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Left: Detailed Ground & First floor plans with pool terrace. Right: 3D exterior concr...

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BIM & STRUCTURAL INTEGRITY SPECS

Structural concrete frame, floor-to-ceiling double-glazed curtain walls, sun sha...



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Signature luxury villa layout featuring master suite wings, outdoor lounge, and garde...

Designing with no limits is every architect's dream, and when it's combined with nature, like this project, it turns to something special. Our designers tried to do their best in creating something unique yet simple to combine with its surrounding

Open space and Semi-open space are used in this project to create different experiences for the user. They can be used in different weather conditions to enjoy the beautiful natural surroundings of the project



BIM & STRUCTURAL INTEGRITY SPECS

BIM LOD 350 architectural model, stone facade cladding schedules, daylight optim...



The combination of concrete and glass materials is like a surreal experience where you can imagine the concrete blocks are floating in the air. The outside pool is a great feature for the villa and can enhance the air ventilation and temperature in the house. It also provides a great playing area for the kids

The most attractive rooms in our homes are usually rooms with the largest openings. We believe that during the daytime the sunshine can improve the bedrooms atmosphere, but it's very important to consider right curtains to be used during the night or when you need darkness to prevent excess light from the outside and thick cream color curtains will properly do same in this project. There's something so cheering and fascinating about floor to ceiling bedroom windows

As you can see in the architectural rendering outcomes the terrace has a special design and, combined with its nature, instead of cutting a tree that was within construction area, we decided to combine it with a terrace and use it as a natural shade

Proper lighting is also completing this beauty, an organic shape chandelier in the middle of the space for the main light and spotlight in the ceiling for focused and area lights



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Comprehensive execution documentation displaying physical on-site steel/concrete erection...

BIM & STRUCTURAL INTEGRITY SPECS

Active site supervision, structural beam-column verification, daylight analysis ...



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ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Left: Hillside stone chalet nestled in mountain slopes. Right: Active on-site concret...

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BIM & STRUCTURAL INTEGRITY SPECS

Rubble masonry retaining walls, timber roof trusses, seismic sloped foundation c...



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ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Villa retreat with private pool and expansive gardens, verified by foundation and poo...

BIM & STRUCTURAL INTEGRITY SPECS

Reinforced concrete pool shell, hydraulic MEP coordination, integrated landscape...



05

JABAAN OFFICE PROJECT

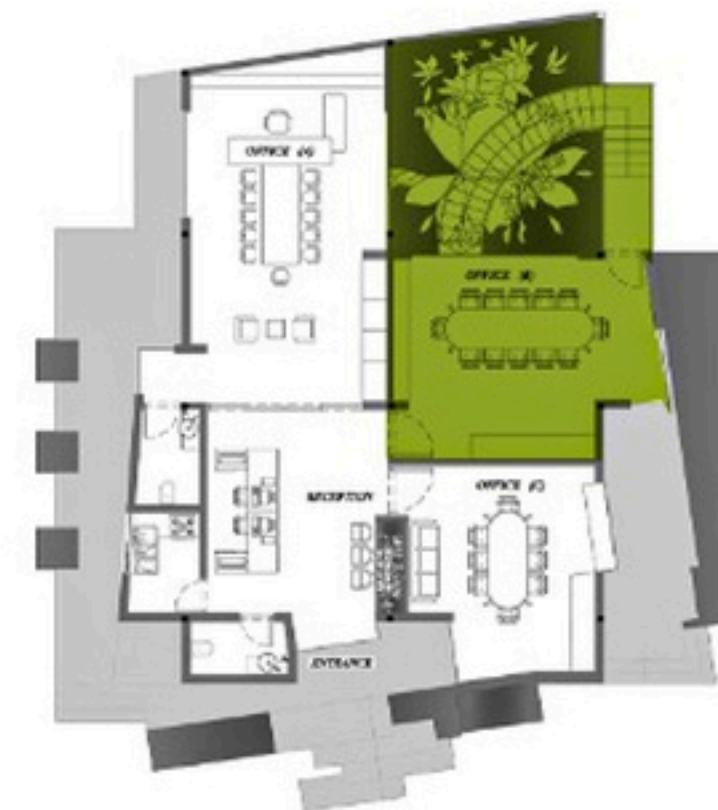


*FINAL ALTERNATIVE
AREA 140_m*

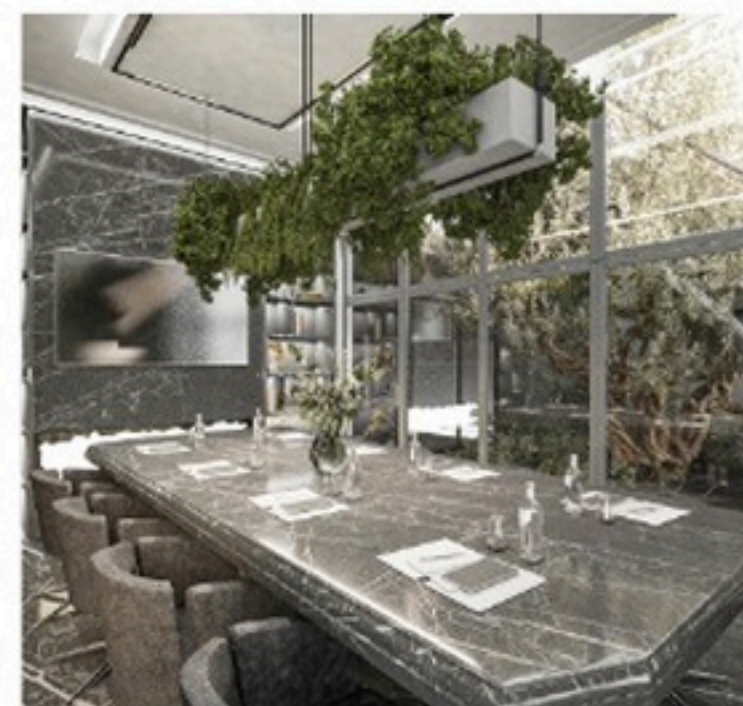
that play off the traditional indoor office space are the ideal. Fortunately, contemporary landscape design techniques and still-evolving technologies offer a myriad of possibilities for organizations to transform exterior spaces into places for employees to take a break and socialize as well as be productive through the workday.

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Modular cubic villas featuring white cantilever frames, dark volcanic stone, and Japa...



*FINAL ALTERNATIVE
AREA 140_m*



BIM & STRUCTURAL INTEGRITY SPECS

Modular structural bay spacing, cantilevered terrace engineering, drip irrigatio...



villa in Australia

Designing with no limits is every architect's dream, and when it's combined with nature, like this project, it turns to something special. Our designers tried to do their best in creating something unique yet simple to combine with its surrounding



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Modern villa in Australia featuring fluid curvilinear balconies, open-plan great room...

Open space and Semi-open space are used in this project to create different experiences for the user. They can be used in different weather conditions to enjoy the beautiful natural surroundings of the project

To enhance the interior and exterior connection, floor to ceiling windows are selected. These windows bring the breathtaking outside natural environment inside the house. The windows frames are in harmony with tree trunks, as they merge with the vertical pattern of trunks



BIM & STRUCTURAL INTEGRITY SPECS

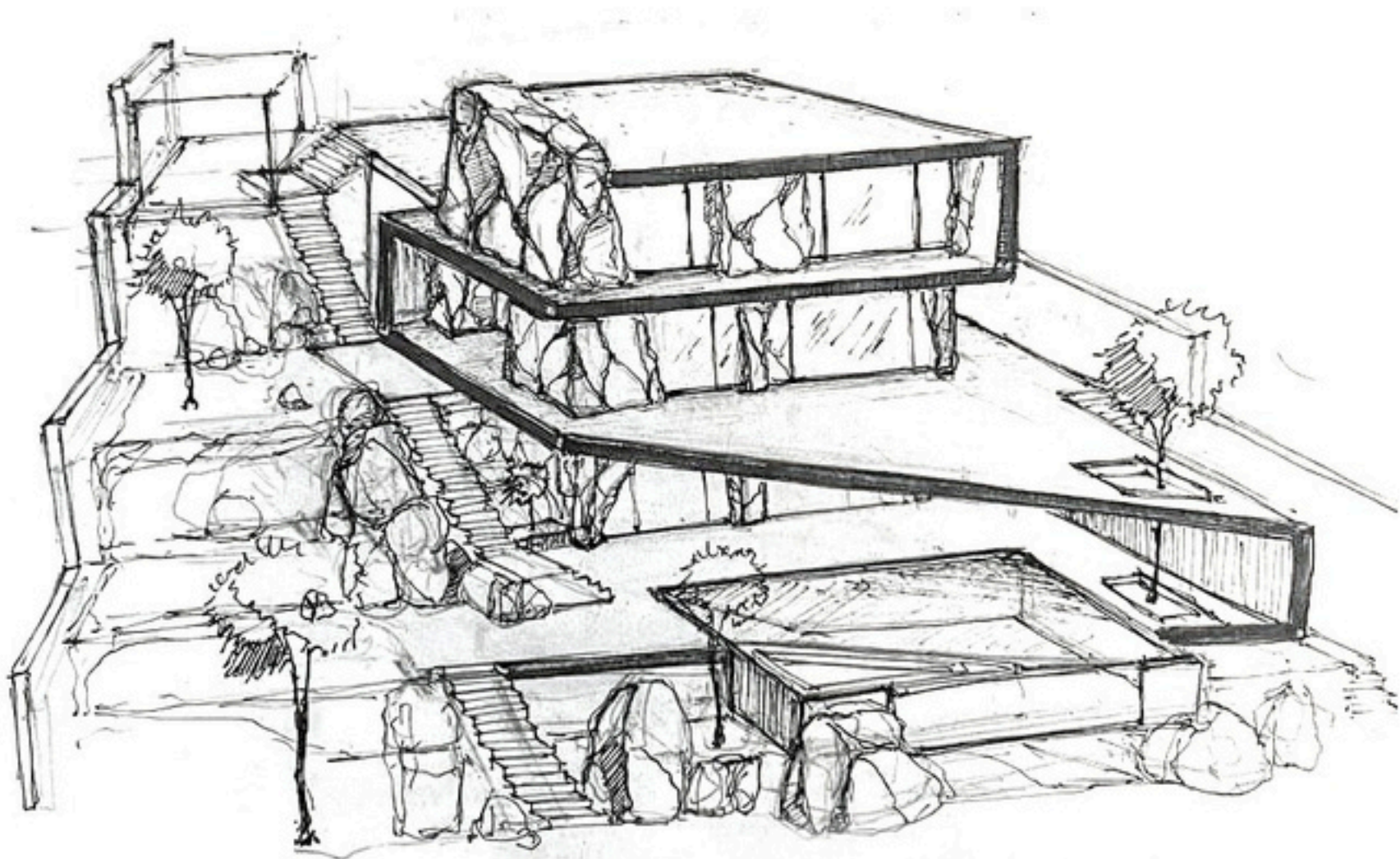
Curved formwork engineering, passive solar shading, luxury master suite layout.



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ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Left: Architectural floor plans & cantilever details. Right: Daring board-formed cant...



BIM & STRUCTURAL INTEGRITY SPECS

Post-tensioned concrete cantilevers, structural glass balustrades, geotechnical ...



URBAN SPACE WISDOM DESIGN
THE EXTERIOR DESIGN
MICRO DESIGN OF URBAN SPACES
LOCATION: PROPHET STREET, ASHRAFI ESFAHANI ST.
YEAR: 2015



Creative City is a concept that by entering the component of creativity in the design, planning and management of the city, a new way for all the people of the society to provide a response to the transcendent needs of human beings such as self-realization, beauty and so on with a strong presence in the public areas of the city, while promoting the sense of satisfaction and civic participation in the quality of the urban space. In fact, creative urban space is a suitable platform that not only accommodates citizens but also is a confluence of pure cultural and artistic thoughts and events.



The wisdom of spaces is not urban angles and dead urban spaces, but on the contrary, because of its smallness, it creates good opportunities to experience hundreds of creative and innovative ways and encourage innovative designers, as well as an opportunity to meet the need for diversity and aesthetic productivity of the city. Considering the importance of shopping activity and the role of walking in the quality of life of citizens (in contrast to the presence of cars and rider movements that disrupt safety and security) in the form of a concept called pedestrian shopping mall, an effective relationship between shopping activity, safety and vitality can be found.



Ambient perception, sense of vision is more involved than other five senses. One of the elements of visual perception is light. Due to the difference between readable and beautiful cities day and night, it can be wrote that the purpose of lighting and beautifying urban spaces is to encourage people to have a stable presence in the space, organize the wall and make the city more attractive, especially at night.





DIAMOND

VILLA TOWN

شهرک دیاموند دارای 60 قطعه زمین به مساحت 600 تا 1500 متر می باشد

حداقل عرض هر قطعه 20 و طول آن 30 متر می باشد

راه اصلی 16 متر و راه های فرعی 9 متری می باشد

تمامی راه ها به طول کلی 17000 متر مربع سنگ فرش می باشد

دارای 5000 متر فضای پارک و تفریحی در 9 پارک مجزا از 200 متر تا 1700 متر و در مجموع 10 هزار اصله درخت در این مجموعه ها می باشد

زمین تئیس زمین چند منظوره والیبال فوتسال بسکتبال کافه بدنسازی سالن تفریحات ورزشی از امکانات جانبی این شهرک می باشد

تمامی تأسیسات برق آب گاز از زیر زمین احداث شده است و دارای کانال جمع آوری آب های سطحی می باشد

در مجموع این مجموعه در 60 قطعه طراحی شده است و با قرار گرفتن امکانات کامل فضایی آرام بخش برای ساکنین مجموعه فراهم کرده است



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Masterplanned gated residential enclave along riverfront greenway with monumental ent...



BIM & STRUCTURAL INTEGRITY SPECS

Macro parcel subdivision, storm water retention swales, hierarchical private roa...

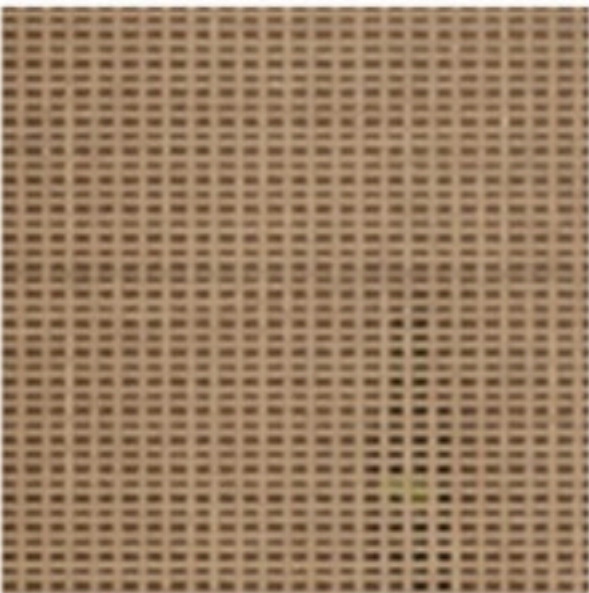


Material Palette



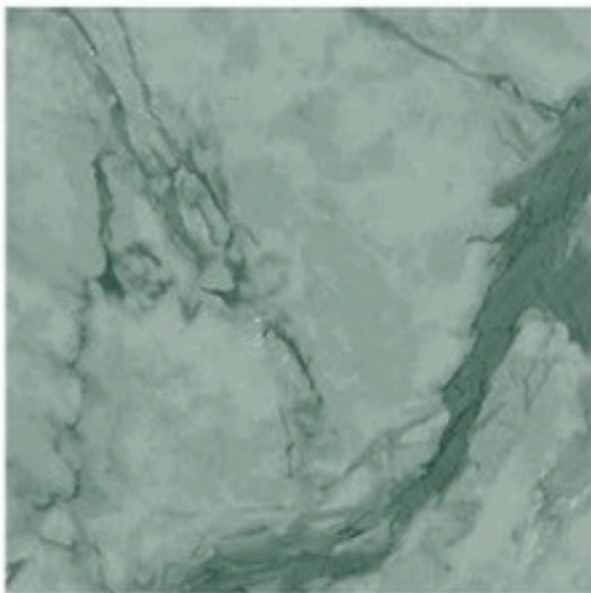
Natural Stone

The term natural stone refers to organic rock quarried from the earth for use as building material. It is a solid composition made up of one or more minerals formed thousands of years ago.



Wicker

Wicker is the oldest furniture making method known to history, dating as far back as 5,000 years ago. It was first documented in ancient Egypt using pliable plant material, but in modern times it is made from any pliable, easily woven material.



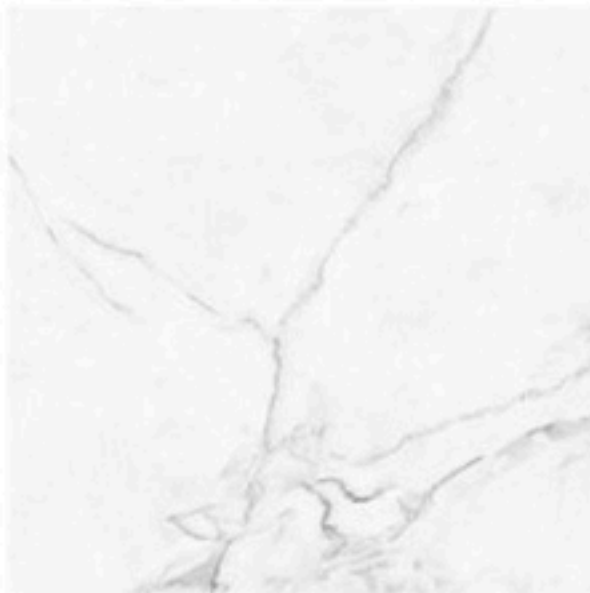
Green Marble

Decisive, ground-breaking and aiming to draw the gaze of every eye. For restless minds who understand that difference is a value in itself.



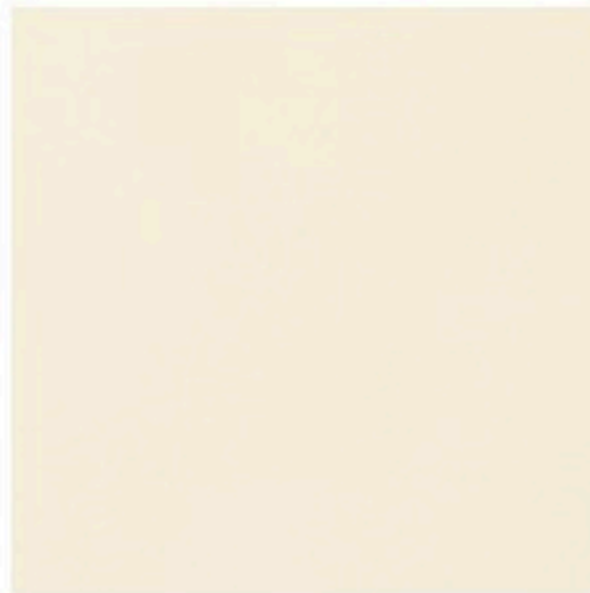
Ulmus Wood

Elms are deciduous and semi-deciduous trees comprising the genus Ulmus in the family Ulmaceae. They are distributed over most of the Northern Hemisphere, inhabiting the temperate and tropical-montane regions of



Marble

Marble is a rock resulting from metamorphism of sedimentary carbonate rocks, most commonly limestone or dolomite (rock).



Spanish MDF Sheet

MDF is traditionally made of wood fibers that have been broken down into residuals, then mixed with wax and resins and heat pressed into dense, durable sheets. A sheet of MDF will be heavy, smooth and even, with a flat surface that can be painted or sealed.

BIMCO PROTOCOL: DESIGN & MODELING

Stages 01 – 03 • Luxury Residential Protocol

STAGE 1: TOPOGRAPHIC & SOLAR ANALYSIS

High-precision drone survey, terrain slope meshing, daylight & thermal calculations.

STAGE 2: SPATIAL CONCEPT & 3D MASSING

Volumetric studies, private vs. entertainment zones, pool & garden integration.

STAGE 3: BIM LOD 350 RESIDENTIAL MODEL

Parametric Revit model, clash detection with structural concrete & MEP piping.

BIMCO PROTOCOL: VISUALIZATION & BUILD

Stages 04 – 06 • From CGI to On-Site Reality

STAGE 4: ULTRA-HIGH-END CGI & LIGHTING

Cinematic renders in Corona/V-Ray, material mockups, interior ambience simulation.

STAGE 5: FULL EXECUTIVE DRAWING SET

1:20 joinery shop drawings, stone cladding details, door/window schedules.

STAGE 6: ON-SITE QUALITY ASSURANCE

Periodic physical site inspections, contractor coordination, build integrity control.

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

BIMCO Studio Barcelona mandates complete continuity from parametric BIM LOD 350 to si...

BIM & STRUCTURAL INTEGRITY SPECS

ISO 19650 BIM Standards, Autodesk Revit LOD 350, Clash Zero Tolerance, Worksite ...

SOHEIL MASTI

BIM Director & Computational Architecture Lead

BIMCO Studio Barcelona • Spain & International

BIM MANAGEMENT & INFORMATION ARCHITECTURE:

- Strategic BIM implementation, Revit LOD 350 parametric modeling
- Multi-discipline clash audits between architectural form, structural frames, and MEP systems
- Automated quantity takeoffs and material schedules for bespoke builds
- International project delivery across Spain, Europe & Middle East
- Digital construction workflows ensuring zero discrepancies on site

SIAVASH PAZOOKI

Senior Architectural Designer & High-End 3D Visualizer

M.Arch & B.Arch • BIMCO Studio Barcelona Partner

CONCEPT DESIGN & VISUALIZATION LEADERSHIP:

- Lead Architectural Concept Designer for signature luxury commissions & masterplans
- Photorealistic architectural rendering, natural lighting moods & cinematic camera angles
- Topographical terrain modeling, landscape synthesis, and material detailing
- Turnkey executive shop drawings, joinery specifications & on-site build supervision
- 8+ years leading award-winning commissions across residential & urban typologies

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

BIMCO Studio Barcelona unites high-level BIM information architecture with signature ...

BIM & STRUCTURAL INTEGRITY SPECS

Registered Architects, BIM Managers (Autodesk Certified), LOD 350 Audit Speciali...



BIMCO

VOLUME I: LUXURY VILLAS & RESIDENCES

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